

SaiVibhavChirravuri

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EDUCATION

University of Southern California - Los Angeles, CA

Master's in Computer Science (GPA: 3.5 / 4.0)

June 2024 - December 2025

SRM Institute of Science and Technology, Amaravathi campus - Amaravathi, India

Bachelor's of Technology in Computer Science & Engineering (GPA: 8.7 / 10.0)

August 2018 - May 2022

WORK EXPERIENCE

University of Southern California

Los Angeles, CA

Research Assistant

September 2024 - December 2024

- Achieved **92.4% validation accuracy** in detecting corneal diseases by fine-tuning **EfficientNet-B3** and **U-Net** models on a curated **10,000+ OCT image dataset**, leveraging **CLAHE**, elastic transformations, and **Dice loss** to boost generalization.
- Accelerated segmentation inference by 4× through **ONNX Runtime** conversion and algorithmic optimization, facilitating near-real-time feedback for clinical collaborators at Keck School of Medicine.
- Engineered a scalable image preprocessing pipeline with **Dask**, **OpenCV**, and **PyTorch Lightning**, executing 500K+ transformations, and tracked experiments via Weights & Biases and TensorBoard for reproducibility.

Amadeus

Bengaluru, India

Software Engineer

August 2022 - May 2024

- Initiated and led modernization of legacy C++ monoliths into scalable Dockerized microservices on **AKS**, driving a 40% cut in deployment time through automated Jenkins pipelines and Helm-based rollout strategies.
- Architected a multi-threaded async task pipeline deploying **RabbitMQ** to offload backend workloads from real-time C++ APIs, cutting customer-facing latency by 30%, and optimized overnight loyalty data processing with Bash automation and MySQL tuning, decreasing runtime by 50%.
- Spearheaded design of a dynamic reward engine in C++, introducing rule-based filtering and performance-tuned MySQL procedures to process 1M+ monthly transactions, and proactively drove cross-team adoption across India, France, and Spain engineering hubs.
- Refactored legacy XML configuration workflows into a dynamic metadata-driven engine maneuvering C++ and XPath parsing, allowing real-time updates to business rules without redeployments and diminishing manual config errors by 80%.

Amadeus

Bengaluru, India

Software Engineer Intern

February 2022 - July 2022

- Took lead in identifying observability gaps and monotonously built a centralized monitoring framework from scratch, integrating custom probes and log aggregation to reduce incident triage time by 60% across 5 services.
- Developed automated service recovery scripts in Python and Bash for handling DB crashes and memory leaks; consolidated them with Jenkins and Azure alert triggers, achieving **<2-minute recovery** in 90% of failure scenarios.
- Containerized internal tools and CI utilities on Docker, allowing uniform builds across environments and simplifying developer onboarding with prebuilt dev images.

PROJECTS

InfraML – (Python, FastAPI, AWS Lambda, Redis, Kafka)

- Developed scalable ML inference pipeline handling 10,000+ async requests/day via Kafka and Redis queues.
- Integrated auto-scaling with AWS Lambda and combined real-time dashboard using Grafana.

CodeAtlas – (React, Node.js, PostgreSQL, Neo4j)

- Created a developer knowledge graph to visualize relationships among 10K+ GitHub projects using Neo4j.

6th Edition Kernel – (C, Assembly)

- Engineered a custom x86 kernel supporting process management, memory paging, and hardware I/O.
- Implemented core system calls and interrupt routines for hardware integration.

TECHNICAL SKILLS

Languages: Python, C++, Java, Bash, C, JavaScript, C#

Frameworks/Tools: Flask, FastAPI, React, Node.js, Express, Jenkins, Docker, Git, Redis, Kafka, gRPC, Scikit-learn, PyTorch, TensorFlow, AWS (EC2, S3, Lambda), Linux, REST APIs, Cronjobs

Databases: MySQL, MongoDB, PostgreSQL